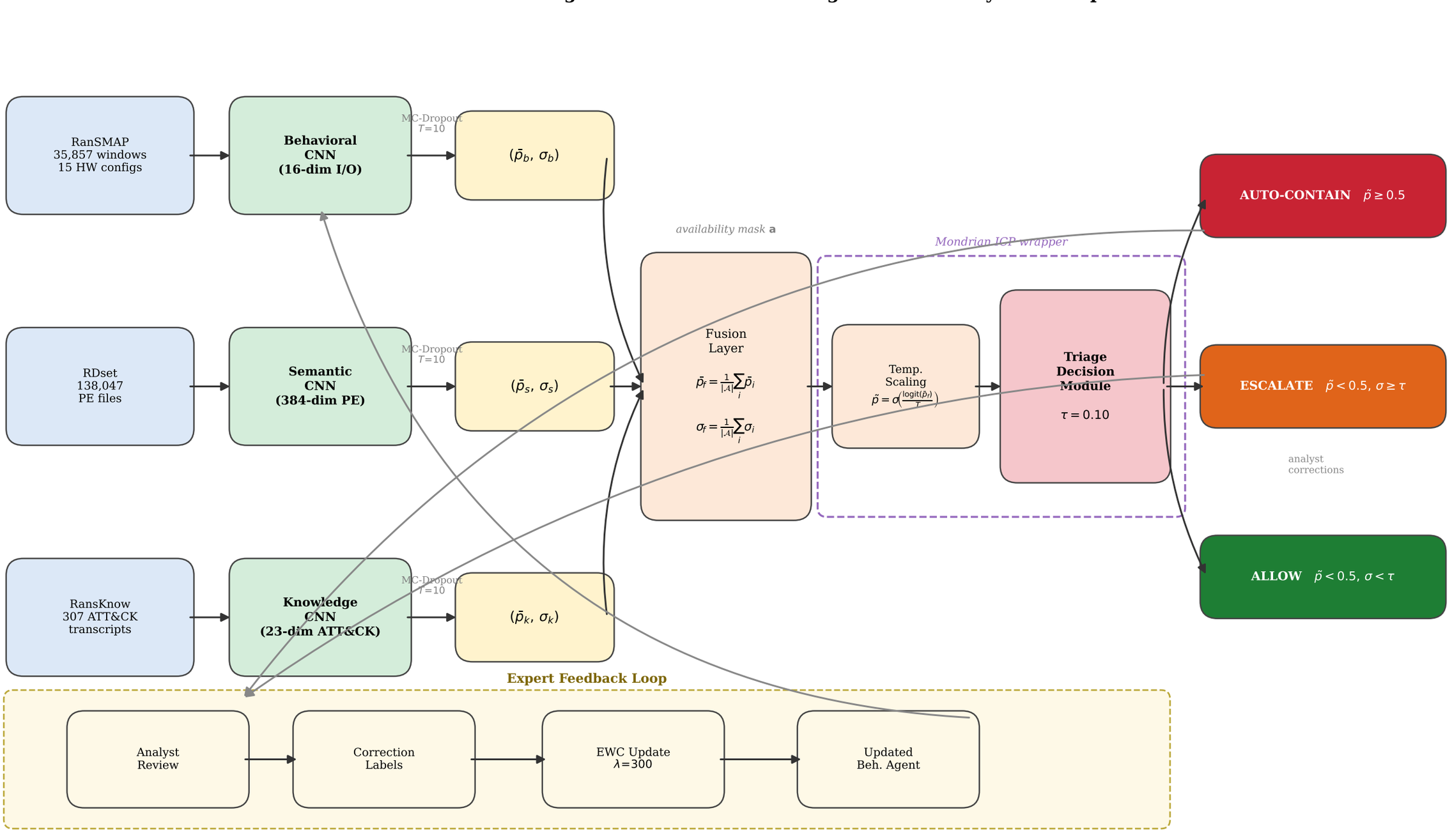

Extended Agentic SABRE+ — Three-Agent Uncertainty-Aware Pipeline
RanSMAP
35,857 windows
15 HW configs
Behavioral
CNN
(16-dim I/O)
MC-Dropout
T=10
$(\bar{p}_b, \sigma_b)$
RDset
138,047
PE files
Semantic
CNN
(384-dim PE)
MC-Dropout
T=10
$(\bar{p}_s, \sigma_s)$
RansKnow
307 ATT&CK
transcripts
Knowledge
CNN
(23-dim ATT&CK)
MC-Dropout
T=10
$(\bar{p}_k, \sigma_k)$
availability mask a
Fusion
Layer
$\bar{p}_f = \frac{1}{|\mathcal{A}|}\sum_i \bar{p}_i$
$\sigma_f = \frac{1}{|\mathcal{A}|}\sum_i \sigma_i$
Mondrian ICP wrapper
Temp.
Scaling
$\tilde{p} = \sigma\left(\frac{\text{logit}(\bar{p}_f)}{T}\right)$
Triage
Decision
Module
$\tau = 0.10$
AUTO-CONTAIN $\tilde{p} \geq 0.5$
ESCALATE $\tilde{p} < 0.5, \sigma \geq \tau$
ALLOW $\tilde{p} < 0.5, \sigma < \tau$
analyst
corrections
Expert Feedback Loop
Analyst
Review
Correction
Labels
EWC Update
λ=300
Updated
Beh. Agent